MAT 5173 — Weekly Schedule

Algebra I · Fall 2026 · UT San Antonio

Texts

- **Rosebrock** — Rosebrock, *Visual Group Theory*
- **Knapp** — Knapp, *Basic Algebra*
- **Beardon** — Beardon, *Algebra and Geometry*
- **CLO** — Cox, Little & O'Shea, *Ideals, Varieties, and Algorithms*
- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*
- **Smith** — Smith, Kahanpaa, Kekalainen & Traves, *An Invitation to Algebraic Geometry*
- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

Schedule

Date		Topic	Reading	Due
Wed 19 Aug		Course launch; the plane and its isometries	Rosebrock Ch. 1	
Mon 24 Aug		Reflections generate everything	Rosebrock Ch. 1	
Wed 26 Aug		From symmetry to the group axioms	Rosebrock Ch. 2	
Mon 31 Aug		Cyclic and dihedral groups	Rosebrock Ch. 2	Problem Set 1
Wed 2 Sep	**Q&A**	Defense 1		
Wed 9 Sep		Permutations and the sign homomorphism	Beardon §1.3-1.4	
Mon 14 Sep		Subgroups; Cayley graphs	Rosebrock Ch. 3	Problem Set 2
Wed 16 Sep	**Q&A**	Defense 2		
Mon 21 Sep		Cosets and Lagrange	Beardon §12.2-12.3	
Wed 23 Sep		Homomorphisms and quotient groups	Rosebrock Ch. 3	
Mon 28 Sep		Group actions: orbits and stabilizers	Rosebrock Ch. 4	
Wed 30 Sep	**Exam**	Midterm Exam 1 — Segment A		
Mon 5 Oct		Orbit-stabilizer and counting	Rosebrock Ch. 4	
Wed 7 Oct		Symmetries of the polyhedra; finite rotation groups in space	Beardon §14.2-14.3	
Wed 14 Oct		Finite groups; frieze and wallpaper patterns	Rosebrock Ch. 7	
Mon 19 Oct		Mobius transformations as a group action on the sphere	Beardon Ch. 13	Problem Set 3
Wed 21 Oct	**Q&A**	Defense 3		
Mon 26 Oct		Rings and polynomial rings k[x], k[x,y]	Knapp §VIII.1	
Wed 28 Oct		Integral domains and fields of fractions; polynomials in one variable	Knapp §VIII.2	
Mon 2 Nov		Ideals; prime and maximal ideals	Knapp §VIII.3	Problem Set 4

Date		Topic	Reading	Due
Wed 4 Nov	**Q&A**	Defense 4		
Mon 9 Nov		Affine space and affine varieties V(S)	CLO §1.2	
Wed 11 Nov		I(V) and the algebra-geometry dictionary	CLO §1.4	
Mon 16 Nov		The coordinate ring k[x,y]/I(V) — the landing	CLO §1.4	Problem Set 5
Wed 18 Nov	**Q&A**	Defense 5		
Mon 23 Nov		Why C and not R; plane curves — rational, cusp, node	Shafarevich §1.1	
Mon 30 Nov	*Review*	Review — the exam pool		
Wed 2 Dec	**Exam**	Midterm Exam 2 — Segment B weighted		